Hospital Analytics (SQL)

Healthcare analytics project on a normalized hospital database using **PostgreSQL**, covering readmissions, length of stay, occupancy and no-shows.

> All data is synthetic and generated inside the script. No real patient information is used.

Business Questions

- What are total admissions, revenue and average length of stay?
- Which departments and doctors handle the most patients and revenue?
- What is the 30-day readmission rate, overall and by diagnosis?
- How does cost vary by age group, gender and insurance type?
- What is daily bed occupancy?
- Which departments have the highest appointment no-show rates?

Database Schema

Table	Purpose
`departments`	Hospital departments
`doctors`	Doctors linked to departments
`patients`	Demographics and insurance
`admissions`	Admit/discharge dates, diagnosis, bill
`appointments`	Appointment status (Completed, Cancelled, No-Show)

SQL Concepts Used

- Normalized schema with primary keys, foreign keys and `CHECK` constraints
- CTEs and multi-step analysis
- Window functions: `LAG` (readmissions), `RANK`, `NTILE`, running totals
- `ROLLUP` for subtotals, `FILTER` for conditional aggregation
- Date logic: `AGE`, `DATE_TRUNC`, `generate_series`
- Views for dashboards

How to Run

1. Create a PostgreSQL 14+ database.

2. Run `healthcare_analysis.sql`.
3. Connect `vw_admission_summary` to Power BI or Tableau.

Key Findings

Replace with your real results after running the queries.

- 30-day readmission rate: **[ ]%**
- Highest-revenue department: **[ ]**
- Diagnosis with highest readmission rate: **[ ]**
- Department with highest no-show rate: **[ ]%**
- Average length of stay: **[ ]** days

Recommendations

- Follow-up programs for diagnoses with high readmission rates.
- Reminder calls for departments with high no-show rates.
- Use occupancy trends to plan bed capacity.

Tools

PostgreSQL, pgAdmin / psql